Hemophilia

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Introduction

- ☑ Hemophilia is an X-linked hereditary bleeding disorder, characterized by deficiency of factor VIII (hemophilia A) or factor IX (hemophilia B).
- ☑ X-linked disorder, females carry gene, and males are affected.
- ☑ 30% are due to spontaneous mutation.
- ☑ Deficiency results in recurrent bleeds into joints and muscles, leading to hemophilic arthropathy and contractures.
- ☑ Incidence of hemophilia A is one in 5,000 and hemophilia B is one in 30,000 live male births.
- ☑ All races and socioeconomic groups are equally affected.
- ☑ Hemophilia C (factor XI deficiency) generally occurs in one in 100,000 people.

Are All Hemophilia Same?

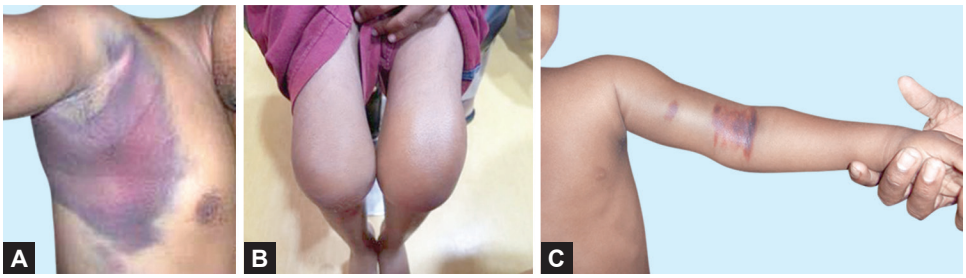
How it varies as per factor level?

Based on factor level in the body, hemophilia A and B is divided as presented in **Table 1**.

TABLE 1: Hemophilia A and B based on factor level.

Severity	***Normal***	***Mild***	***Moderate***	***Severe***
Factor level (VIII/IX)	50–150%	6–50%	1–5%	<1%
Bleeds	NA	Post major trauma, surgery	Post trauma, surgery, occasionally spontaneous	Spontaneous bleeds

Sites of Bleeding (Figs. 1A to C)



Figs. 1A to C: Sites of bleeding.

- ☑ *Joint bleeding (incidence 60–80%)*:
  - Common joints involved are knees, elbows, ankles, shoulders, wrist, and hips
  - Target joint with tendency of recurrent bleeds
  - *Subacute arthropathy*: It develops after repeated bleeds into the joint
  - *Chronic arthropathy*: Progressive destruction of a joint
  - *End stage*: Fibrous joint contracture and disorganization of articular surfaces
- ☑ *Muscle hemorrhage (20–30%)*:
  - *Common muscles*: Iliopsoas, calf, upper arm and forearm, thigh, shoulder area, and buttock
  - *Symptoms*: Pain, swelling, and muscle spasm
  - *Complications*: Nerve compression and contracture
- ☑ Soft tissue—bruising
- ☑ *Life-threatening bleeding (2–5%)*: Intracranial bleeds, intra-abdominal, Iliopsoas, etc.
- ☑ *Other bleeds*: Mouth, nose, scrapes, and minor cuts (5%).

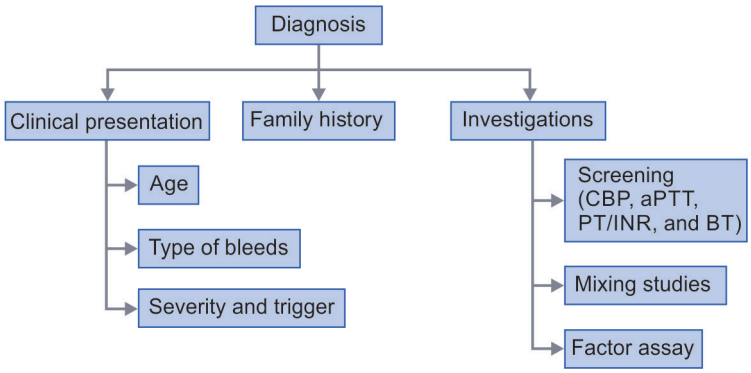
Symptoms of bleed are presented in **Figure 2**.

Fig. 2: Symptoms of bleed.

Sites of Bleeding

Diagnosis of hemophilia is described in **Flowchart 1**.

Flowchart 1: Diagnosis of hemophilia.



(aPTT: activated partial thromboplastin time; BT: bleeding time; CBP: complete blood profiling; INR: international normalized ratio; PT: prothrombin time)

Management

- ☑ A person with hemophilia is born with it and has it all his life.
- ☑ The amount of factor VIII or IX in blood usually stays the same throughout his life (exception factor IX Leiden).
- ☑ With appropriate management, can lead near normal life.

Management

General Measures

- ☑ *Precautions*: Avoid injuries, intramuscular injections, and nonsteroidal anti-inflammatory drugs (NSAIDs)
- ☑ Vaccination to be given subcutaneously
- ☑ First aid treatment for bleeds—**R.I.C.E (Fig. 3)**.

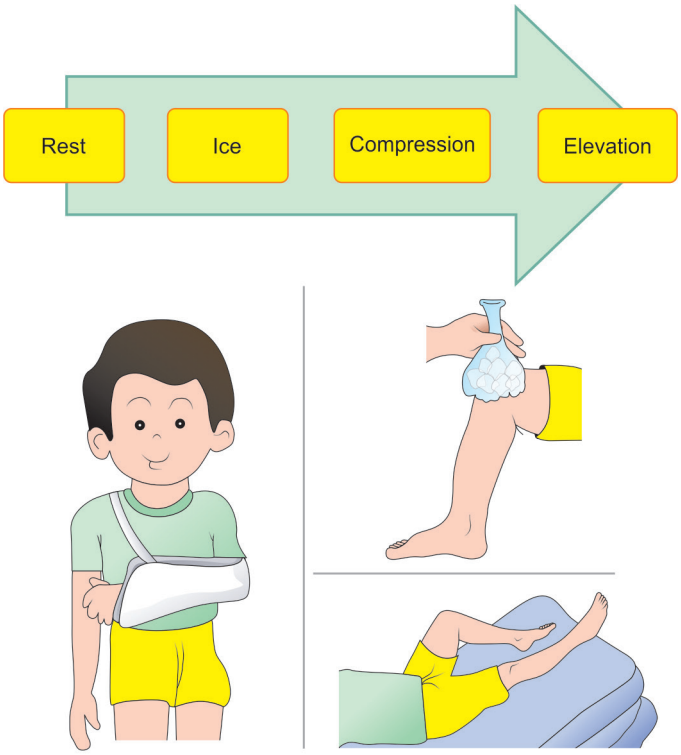


Fig. 3: First aid treatment for bleeds.

Specific Measures

- ☑ For minor bleeds, antifibrinolytics (tranexamic acid) can be used topically or systemically (oral/intravenous).
- ☑ *Desmopressin (DDAVP)*: It boosts plasma level of factor VIII in mild-to-moderate cases. No effect on factor IX.
- ☑ *For major bleeds*: Factor replacement as per bleed **(Table 2)**.

TABLE 2: Factor replacement as per bleed.

Hemorrhage type	***Hemophilia A***		***Hemophilia B***	
	Desired level	***Duration (days)***	***Desired level***	***Duration (days)***
Joint	40–60%	1–2; longer if inadequate response	40–60%	1–2; longer if inadequate response
Muscle (except iliopsoas)	40–60%	2–3; longer if inadequate response	40–60%	2–3; longer if inadequate response
Iliopsoas ☑ Initial ☑ Maintenance	80–100% 30–60%	1–2 3–5 (sometimes longer as secondary prophylaxis during physiotherapy)	60–80% 30–60%	1–2 3–5 (sometimes longer as secondary prophylaxis during physiotherapy)
CNS/head ☑ Initial ☑ Maintenance	80–100% 50%	1–7 8–21	60–80% 30%	1–7 8–21
Throat and neck ☑ Initial ☑ Maintenance	80–100% 50%	1–7 8–14	60–80% 30%	1–7 8–14
Gastrointestinal ☑ Initial ☑ Maintenance	80–100% 50%	1–6 7–14	60–80% 30%	1–6 7–14
Renal	50%	3–5	40%	3–5
Deep laceration	50%	5–7	40%	5–7
Surgery (major) ☑ Preoperative ☑ Postoperative	80–100% 60–80% 40–60% 30–50%	1–3 4–6 7–14	80–100% 60–80% 40–60% 30–50%	1–3 4–6 7–14

(*Source*: Srivastava A, Santagostino E, Dougall A, Kitchen S, Sutherland M, Pipe SW, et al. WFH Guidelines for the Management of Hemophilia, 3rd edition. Haemophilia. 2020:26(Suppl 6): 1-158.)

Prophylaxis

- ☑ Prophylaxis prevents bleeding and joint destruction.
- ☑ Aim of prophylaxis is to keep factor VIII/IX levels >1% all the time.
- ☑ *Principle of prophylaxis*: Convert bleeding pattern of severe hemophilia into milder bleeding pattern of moderate hemophilia.
- ☑ Long-term goals:
  - Improvement in quality of life
  - Reduction in long-term social costs.
- ☑ Prophylaxis can be started at any age.
- ☑ Types of prophylaxis **(Table 3)**.

TABLE 3: Types of prophylaxis.

Primary prophylaxis	Regular continuous treatment initiated in the absence of documented osteochondral joint disease, determined by physical examination and/or imaging studies, and started before the second clinically evident large joint bleed
Secondary prophylaxis	Regular continuous treatment started after two or more bleeds into large joints and before the onset of joint disease documented by physical examination and imaging studies
Tertiary prophylaxis	Regular continuous treatment started after the onset of joint disease documented by physical examination and plain radiographs of the affected
Intermittent prophylaxis	Treatment given to prevent bleeding for periods not exceeding 45 weeks in a year

Inhibitors

- ☑ Inhibitor development is a known complication in hemophilia and refers to immunoglobulin G (IgG) antibodies that neutralize clotting factors.
- ☑ Patients with inhibitors may suffer more complications than a patient without inhibitors such as joint damage.
- ☑ Most inhibitors usually occur within first 50 exposure days of factor replacement therapy.
  - Inhibitor risk is low with regular prophylaxis.

Inhibitors

- ☑ *Inhibitor frequency*:

Type	*Moderate*	*Severe*
Hemophilia A	5–10%	30%
Hemophilia B	5–7%	1.3–3%

- ☑ Confirmation of the presence of an inhibitor and quantification of the titer is performed by Nijmegen modified Bethesda assay.

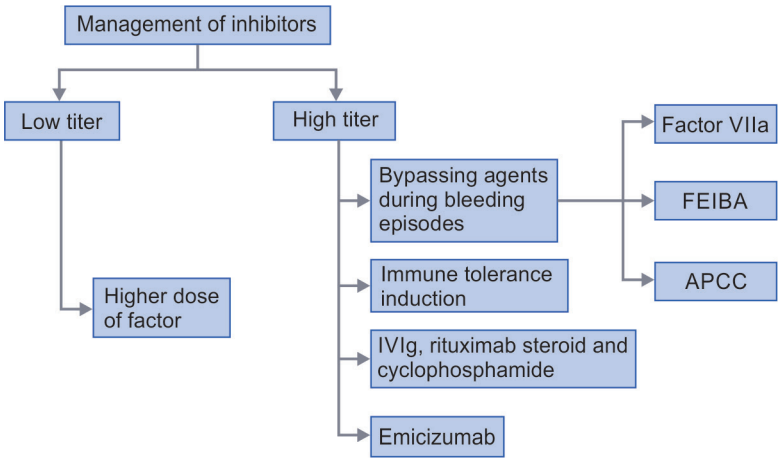
<5 BU	>5 BU
Low responding inhibitor	High responding inhibitor

- ☑ *Screening for inhibitors*: Whenever inadequate response to factor therapy or as a routine below:

Exposure days	*Frequency of screening*
<20	Every 5 exposure days
20–50	Every 10 exposure days
50–150	Twice a year
>150	As and when required

Flowchart 2 describes management of inhibitors.

Flowchart 2: Management of inhibitors.



Management

Management

Emicizumab

Humanized bispecific antibody that binds to both the activated coagulation factor IX and to factor X, mediating the activation of the latter.

- ☑ Prophylaxis for severe hemophilia A patients with or without inhibitors
- ☑ Very effective in preventing break through bleeds
- ☑ Lower dose than recommended in trials is also effective (unpublished Indian data suggestive).

Further Reading

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Preterm with Respiratory Distress

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